Evaluating the Transformative Digital Pedagogies for Higher Education Training Programme: Insights into Participants' Perceptions

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Abstract. The adaptation of education and training systems to the digital age is one of the priorities of the European Commission. Aiming at promoting high-quality, inclusive education, Higher Education (HE) institutions are working towards digital and pedagogical transformation; however, lack of both pedagogical skills for using digital technologies and digital skills for implementing innovative teaching practices make this a challenging task. The Erasmus+ funded Transformative Digital Pedagogies for Higher Education (TDP4HE) project addresses this challenge by developing a training programme for HE staff designed to enhance academic staff's pedagogical and digital competencies, enabling them to effectively transition to digital education. This paper examines the first implementation and evaluation of this online training programme, delivered to EUt + university staff in May-June 2024. The results show participants' positive perceptions about the training programme's innovative approach to transformative digital pedagogies. They also identify certain challenges encountered and reveal ways in which the training programme could be improved to better meet educators' needs.

Keywords: Digital Pedagogies · Online Training Programme · Digital Competence · Higher Education · Programme Evaluation

1 Introduction

Among the most important initiatives of the European Commission is the support of the adaptation of the education and training systems of Member States to the digital age creating opportunities for high-quality, inclusive education for all [1]. For many years now, the development of digital competences of citizens in general and educators in particular has been a priority of the European Commission [2]. This focus stems from the realisation that digital literacy is crucial for navigating the increasingly digital world and participating fully in society. EU's emphasis on digital transformation is manifested in a) the Digital Competence Framework for Citizens (DigComp) [3], which provides a

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common understanding of digital competence, b) the Digital Competence Framework for Educators (DigCompEdu) [2], which focuses on the specific digital competencies that educators need in order to effectively integrate technology into their teaching practices, and c) funding programmes such as Erasmus+ and Horizon 2020 to support initiatives that promote digital skills development among citizens and educators. Furthermore, according to the International Association of Universities [4], digital transformation is among the strategic priorities of Higher Education (HE) institutions accelerated during COVID-19 pandemic.

Nevertheless, there are several challenges to be considered, such as the digital divide that may exist between countries and educational institutions, which prevents equitable access to technology and digital resources. Another challenge is the fact that technology is advancing rapidly, which requires continuous learning and adaptation for both citizens and educators. Last but not least, one of the greatest challenges is the provision of adequate training and professional development opportunities for educators to effectively use digital tools in their teaching. On the one hand, the lack of pedagogical competences linked to the use of digital technologies, and on the other hand, the lack of digital competences linked to the implementation of innovative and transformative teaching practices, make this effort more complicated.

Inspired by these facts, the Transformative Digital Pedagogies for Higher Education (TDP4HE) project, an ERASMUS+ project submitted under the action KA220-HED - Cooperation partnerships in Higher Education [5], proposes a transformative pedagogical model for the development of pedagogical competences of academic staff. One of the major activities of the project is the design and implementation of an online training programme aiming at developing the transformative digital pedagogies of academic teaching staff. The TDP4HE training programme aims to empower academic teaching staff with both pedagogical and digital competences and to increase their capacity and readiness to manage a real shift towards digital education with their teaching practices. This paper focuses on the implementation and evaluation of the training programme, which was offered online to members of the academic teaching staff of the Universities of the European University of Technology alliance (EUt+) between May – June 2024.

2 Literature Review

2.1 Digital Transformation in HE

According to Mahlow and Hediger [6], digital transformation refers to the transformation of organizational processes and building of new competencies and models through digital technologies in a profound and strategic way. In other words, it does not only concern the use of digital technologies, but it also involves a change of culture, and the acquisition of new skills to facilitate the effective utilisation of technology in order to achieve better outcomes. Bygstad et al. [7], use different terms to define digital transformation in HE. First, they refer to “dual digitalisation” which involves a combination of digital tools, such as Learning Management Systems (LMS), MOOCs, course websites, etc., and digital subjects developed by academics. Then they add that, currently, this “dual digitalisation” has been viewed in a more holistic and comprehensive way, leading to a reconsideration of the learning process; the authors refer to the term

"digital learning space", which involves a combination of the technical, pedagogical and organisational perspective.

During the last decades, emphasis has been placed on digital transformation in HE due to various reasons. First of all, the needs and the expectations of digital native students are changing, and employability skills seem to hold a prominent position in academic curricula, equipping students to meet today's challenges in the modern workplace [8]. Moreover, emerging technologies offer new possibilities for teaching, learning and research, making education more accessible and inclusive [9]. Additionally, HE institutions face increasing competition, both domestically and internationally, and digital transformation can help them stand out and excel both in the achievement of better learning outcomes and in research. The need for digital transformation was intensified with the abrupt shift to remote learning during the COVID-19 pandemic, which forced HE institutions to quickly adapt and offer flexible learning options. This primarily involved leveraging digital platforms, adopting innovative teaching methods, and creating new digital learning materials [10].

In the 2030 Agenda for Sustainable Development, adopted by all United Nations Member States in 2015, quality education is one of the fundamental human rights and a cornerstone of the 17 Sustainable Development Goals [11]. Digital transformation is fundamental in the improvement of HE. It must be carefully guided to enhance quality education, while proactively mitigating potential risks and negative impacts [4].

2.2 Transforming Pedagogy in Higher Education

The Digital Education Action Plan (2021–2027) is a renewed EU policy initiative that sets out a common vision of high-quality, inclusive and accessible digital education in Europe; it promotes the development of a high-performing digital education ecosystem and enhances digital skills and competences for digital transformation [1]. Teachers' lack of digital competence and pedagogical knowledge in HE is one of the major barriers to the realisation of this vision. Members of academic staff need to undergo training that will equip them with the digital skills and pedagogical knowledge to effectively integrate technology into their teaching [12]. They also need to be provided with the appropriate support in adapting to this evolving digital landscape.

Over the last couple of decades there has been an increase in literature pertaining to the professional development of lecturers; however, literature that combines professional development of academics and educational innovation with the use of new technologies in the context of HE is scarce [13]. According to Ter Beek et al. [13], educating lecturers to apply and integrate innovative technology in their curricula is essential for the future of education, and it is up to HE institutions to facilitate effective forms of lecturer professional development to scale up the pace of change. A study by Bitar and Davidovich [14] provides an exploration of lecturers' experiences in terms of the Digital Learning initiative implemented by the Council for Higher Education (CHE) and the Planning and Budgeting Committee (PBC) in Israel. The results of the study reveal four key domains that shape lecturers' experiences with Digital Learning: educational, personal, cultural and social, and institutional. These domains interact in complex ways, influencing the adoption and effectiveness of Digital Learning strategies. The study also shows

that effective Digital Learning implementation, apart from the integration of technological, pedagogical, and content knowledge, requires an understanding of the cultural and institutional landscape.

The efforts to digitalise and transform pedagogy in HE takes place along with a shift to more learner-centred and active learning approaches to education [12]. Such approaches place emphasis on student engagement in real-life problems and challenges through activities like group work, discussions, problem-solving, and hands-on projects. According to Santoveña-Casal and López [15], in general, an analysis of the topics and research in the last years reflects the COVID-19 pandemic's impact on HE as well as the need of HE institutions to significantly expand the use of digital pedagogy by adapting their teaching methodologies. This demands a significant increase in research conducted in this area of knowledge. Even though the shift to digital technologies was massive during the pandemic, many academics did not embrace student-centered teaching approaches, and the need for education and support on the pedagogical use of new technologies was strengthened [16].

The TDP4HE project proposes a transformative pedagogical model for the development of pedagogical competences of academic staff, that makes integral use of digital technologies [5]. Research conducted in the project is based on the view that the digitalisation of teaching and learning processes is not simply the application of digital technologies to enhance traditional teaching methods. Digitalisation in classrooms is revolutionary in that it makes it easier to adopt radically different and innovative student-centred approaches. By adopting transformative pedagogical approaches, this project aims to develop a model that links transformative pedagogies (learner-centred, opposed to transmissive, teacher-centred, approaches) to the use of digital technologies and to develop a transdisciplinary training programme to foster innovative ways of teaching and learning for academic teaching staff from different disciplines.

3 The TDP4HE Training Programme

The Training Programme on Transformative Digital Pedagogies for academic teaching staff aims not only at improving teachers' technological competency but also their understanding of how to integrate content, pedagogy and technology together in an innovative and transformative way. More specifically, the programme aims to contribute to the general objectives of the project by training academic teaching staff to understand and apply effective teaching methodologies in HE, recognise technology's role in modern pedagogy, design courses, implement diverse teaching methods, integrate new technologies, employ varied assessment techniques, use digital resources effectively, and adapt pedagogical strategies to evolving educational environment.

Grounded in the literature described in the previous section, the programme was designed based on the results obtained from needs analysis on the pedagogical practices implemented in the eight universities that form the European University of Technology (EUt+). Among the results of the needs analysis was the need to focus on transformative approaches (learner-centred) as opposed to transmissive (teacher-centred) approaches. Furthermore, the main topics foregrounded by needs analysis were summarised: learner-centred approaches, inquiry-based approaches, problem-based learning

and problem-solution skills, project-based learning, case-based learning, critical thinking, social-collaborative learning, inter and transdisciplinary thinking skills, and lifelong learning.

The designed online course training programme on transformative digital pedagogies for academic teaching staff includes five different modules, which were incorporated into the project's learning management system (LMS: Moodle). Delivered fully online, the programme features synchronous webinars with pre-tasks, sessions, and assessments. Teaching methods include lectures, demonstrations, small group collaboration, discussions, and assigned readings. The five modules cover teaching in the digital era, using emerging technologies, study process principles, psychology, inclusive education, and academic staff development. Participants are expected to actively engage in theoretical discussions and hands-on activities, integrating theory with practice through evaluated tasks. The first, pilot phase of the implementation of the training programme took place online in Fall 2024 with academics from the European University of Technology (EUt+); the participants included teaching staff from different departments of the project partner universities and associate partners.

A trainer's guide is also planned as a deliverable from the Training Programme on Transformative Digital Pedagogies. This will focus on the content and methodology of the training programme for academic teaching staff and will be available in the project's Open Community of Practice and the project website.

Continuous evaluation of training programs like the TDP4HE is essential to ensure they remain effective, relevant, and aligned with the evolving needs of educators and learners. By systematically assessing the programme's design, delivery, and outcomes, stakeholders can identify areas for improvement and ensure that the training equips educators with the necessary skills to adapt to technological advancements and pedagogical shifts [17]. Continuous evaluation was a key component in the design of the TDP4HE programme as a way of fostering a culture of accountability and reflective practice, enabling participants and shareholders to provide feedback and share insights that can refine the programme's content and methodologies. Moreover, as digital tools and teaching strategies evolve rapidly, continuous assessment helps maintain the program's relevance, ensuring that it consistently supports educators in fostering engaging and transformative learning experiences in higher education [18].

Incorporating principles of design-based research (DBR) into the evaluation process allows for iterative testing and refinement of the program in real-world contexts, ensuring that it responds dynamically to practical challenges and evolving educational environments [19].

4 Methodology

This study is based on the principles of Design-based research (DBR), an interventionist research method [20]. DBR is a methodology designed by and for educators that seeks to increase the impact, transfer and translation of education research into improved practice [21]. In addition, it stresses the need for theory building and the development of design principles that guide, inform and improve both practice and research in educational contexts [22]. As already mentioned, the design of the training programme described

in this paper reflected the outcome of the needs analysis and is in the process of being refined through two cycles of implementation, data collection, analysis and revision. DBR follows iterative cycles of (1) problem identification and definition, (2) context analysis, (3) design and development, (4) implementation, (5) evaluation, (6) reflection and redesign, and (7) communication and dissemination. Phases (1) to (4) of this study were reported in Soule and Kakoulli Constantinou's [23] paper. The present paper focuses on phases (5) to (7), aiming at presenting the way the TDP4HE training programme was evaluated by the participants, discussing their reflections and suggestions for the redesign of the programme and last but not least, communicating the results to different experts and stakeholders interested in transformative digital pedagogy in the HE context.

4.1 Research Questions

The research questions underpinning the study were the following:

1. What are participants' perceptions about the TDP4HE training programme?
2. What are the challenges that participants faced during the implementation of the programme?
3. What are participants' suggestions concerning improvements that need to be made?

4.2 Research Tools

Both qualitative and quantitative data were collected to evaluate the TDP4HE training programme and explore participants' perceptions. The data were collected through a focus group discussion and an online questionnaire which were used to highlight strengths and identify improvement areas. The focus group session lasted 45 min and was video-recorded and transcribed for analysis. The focus group included 10 questions designed to gather detailed participant feedback. The questions were:

1. What are your initial thoughts about the overall experience of this training programme?
2. How do you feel the training programme has impacted your knowledge/understanding of teaching methodologies?
3. Has this programme helped you understand how important technology is in contemporary pedagogy in HE?
4. How confident are you in your ability to formulate learning objectives, design course content, and establish learning outcomes after attending the training programme?
5. Do you feel the programme objectives align with your personal and professional development goals?
6. How much do you think that this programme has helped you become familiar with some learner-centred approaches and teaching methodologies like project-based learning, social collaborative learning, etc.?
7. How much do you think the programme has helped you with integrating emerging technologies into your teaching practices?
8. How do you expect this programme to impact your teaching practices in the short term and long term?

9. How do you feel about the structure and content of the TDP4HE training programme? Do you have any suggestions for improving the programme to better meet your needs?
10. Any final comments/ thoughts?

The data from the focus group were analysed using thematic analysis, which involved manually identifying and categorising key themes and patterns within the responses. This process entailed a careful review of the focus group transcripts, allowing for an in-depth understanding of recurring ideas and insights. By organising the data into distinct themes, we were able to draw meaningful conclusions and gain a comprehensive view of the participants' feedback.

The online questionnaire included 12 questions designed to gather detailed feedback from the participants. In particular, questions 1, 2 and 12 were open-ended and aimed at prompting participants to elaborate on their experience (e.g. what they liked the most/least and suggestions for improvements. Questions 3 to 11 were five-point Likert statements measuring participants' agreement (Strongly Disagree-Strongly agree). All questions are listed below:

1. What was the one aspect you found most interesting or useful in the TDP4HE Training programme?
2. What was the one thing you liked the least, or which you would change about in the TDP4HE Training programme?
3. The TDP4HE Training programme contents met my expectations.
4. The content has been clear and understandable.
5. The LMS/ Moodle platform worked well.
6. The synchronous sessions were organised to allow all participants to fully participate.
7. The pre- and post-activities were useful.
8. The asynchronous materials were useful.
9. The TDP4HE modules were useful, so that I am more confident about its topic now.
10. I think this training programme is useful for my professional/academic training.
11. I would recommend this training programme to other colleagues.
12. Please provide any other comment, suggestion or feedback about the training programme that was not covered in the questions above.

The data from the online questionnaire were analysed manually using participants' written responses to the open-ended questions, and the graphs were generated automatically using Google Forms. Both research tools allowed gaining a comprehensive understanding of their feedback.

4.3 Participants

A total of 106 members of the academic staff of all nine Universities of the EUt + alliance registered in the course. Participants represented Riga Technical University, University of Technology of Troyes, Cyprus University of Technology, Technical University of Cartagena, Technological University Dublin, Technical University of Sofia, Technical University of Cluj-Napoca, Darmstadt University of Applied Sciences, and The University of Cassino and Southern Lazio. Almost half of the 106 academics participated actively in the synchronous sessions. Twenty-four academics across the EUt+

Alliance completed the online questionnaire. In addition, four academic staff members from the EUt + Alliance who attended the TDP4HE Training Programme sessions participated in the focus group conducted by two interviewers from the Cyprus University of Technology.

5 Results

The findings of the focus group discussions and online questionnaire provide insights into participants' perceptions of the TDP4HE training programme highlighting its strengths, challenges, and areas for improvement.

The findings section presents eight key thematic areas, each illustrated with relevant excerpts from focus group discussions and visual representations from the online questionnaire, such as graphs, to enhance clarity and depth of analysis. By integrating qualitative and quantitative data, this section provides a holistic understanding of the programme's impact and effectiveness.

5.1 Interesting/Useful Aspects of the Programme

In the focus group discussion, participants expressed initial curiosity and motivation to improve their teaching methods and stay updated with the latest advancements in the field. This was driven by a desire to find their place within a larger network of professionals and specialists.

The following extracts illustrate this theme:

So I have to, I must admit that my initial thoughts were most of curiosity and how to find my own connection, to find my own spot into this large network of professionals, of specialists in the field. (Participant 1)

So, once again, I, you gave me the confidence and you're-motivated me that I am on the right track, according to the methodologies I use with my students. (Participant 1)

So it made us, made me at least, consider differently how to use technology in the classroom and how the lesson can be more interactive and that students could gain more motivation when using these technologies in classroom. (Participant 3)

Also, the question, "*What was the one aspect you found most interesting or useful in the TDP4HE Training programme?*", shows as the most interesting or useful the diverse trainers' backgrounds, the selected topics and materials presented utilising theoretical and practical means at a great level, the communication between them and exchange of ideas. Participants' responses are also shown in Table 1 below:

Furthermore, participants found the asynchronous materials useful, as 91.6% strongly agreed or agreed with the statement, while a few remained neutral or disagreed (Fig. 1).

Participants also think that the training programme is useful for their professional/academic training, as 91.7% of them strongly agreed or agreed with the statement, while a few remained neutral (Fig. 2).

Table 1. Interesting and useful aspects of the training programme

• The internationalisation aspect of the team of trainers from different universities/countries in EUT+. • The diversity of the selected topics, the up-to-date references, materials and cases presented. • The communication between the participants and the opportunity to share experiences. • The group discussions and interactions they had in the break-out rooms. • The multifaceted approach to the problem was a skillful combination of theoretical background and interactive, hands-on activities and practical tools. • The innovative teaching and learning techniques/ teaching through new/ emerging technologies. • The modules' content particularly Module 2, "Didactic Foundation of Learning/Teaching through the use of new/ emerging technologies".

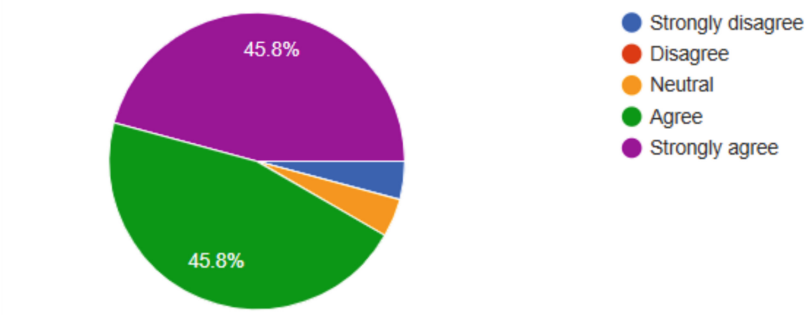


Fig. 1. The asynchronous materials were useful

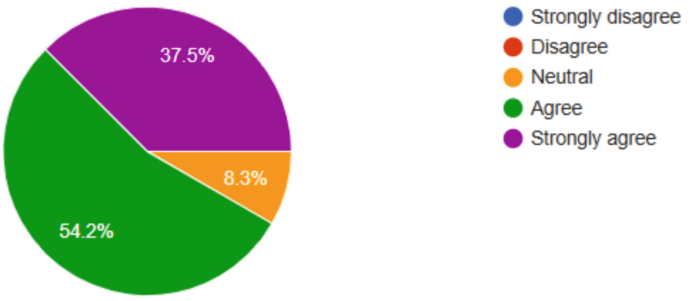


Fig. 2. This training programme is useful for my professional/academic training

5.2 Things that Participants Liked/Didn't Like

Participants liked the fact that the programme reassured them about the validity of their existing teaching methodologies, such as project-based learning and student-centred approaches. They appreciated knowing that their methods aligned with contemporary practices.

The excerpts below showcase this theme:

I came across your course and I am thankful to that it's actually that you highlighted the fact that I knew it was the right way to do it. It is the right way to do it. So, project-based learning, student-centered approach, and technologies to be used in the delivery, in the teaching and the learning and the assessment part of our students should be incorporated and not neglected. (Participant 1)

Yeah..It's actually, it was very reassuring and very comfortable, you know, to know that what you do, you do it in the right way. Because there are episodes, moments in our careers when we question our approach, our teaching methodologies as a result of the experience, since we have got or with the discussions. (Participant 1)

Another aspect that participants highly valued was the programme's structure, which involved collaboration with trainers from different universities. Participants appreciated the opportunity to interact with colleagues from various disciplines and share diverse perspectives:

So it was convenient for us. And I also really liked that we had the chance, it was interactive, we had the chance to speak about our minds, we had the breakout rooms, so we were meeting with the other colleagues and we had to discuss about certain things. So it was overall a very fruitful and thoughtful experience full of new knowledge. (Participant 3)

On the other hand, the question "What was the one thing you liked the least, or which you would change about in the TDP4HE Training programme?" revealed the least favourite parts of the training programme and/or suggestions mainly focused on the overall organisation, training content and teaching approach. A few of them stated that nothing should be changed (Table 2).

Participants also considered the time allocated for each breakout room insufficient. They suggested extending these sessions to allow for more thorough discussions and exchange of ideas.

Talking about the length of the breakout rooms, you know, because you try to keep the balance between the theory and the presentations and then the interaction in the breakout rooms. And it worked as it worked, but we felt like, and I'm sure you heard the complaint, it was too little time. We needed more time to, you know, maybe to expand a little bit the timing for how the breakout rooms work next time you organize a new program, right, to provide each and every speaker who is willing to share their experiences the amount of time they need to do that. (Participant 1)

5.3 Meeting Participants' Expectations

Regarding whether the TDP4HE Training programme contents met their expectations, as Fig. 3 illustrates, 75% of participants strongly agreed or agreed with the statement, while a few remained neutrals, disagreed or strongly disagreed.

Table 2. The least favourite parts of the training programme

Category	Specific points that could change
Overall organisation	• Follow-up emails from different persons made it difficult to retrieve the information sent • In some cases, long synchronous sessions coincided with personal/family time; they could be kept only for questions • A workshop could be organised in person to complete the programme
Training content	• Presentations could be less theoretical and more interactive, with more practical aspects, guidelines, examples, and best practices guidelines for teaching and implementing relevant pedagogical theories • Some sessions were at a basic level • Some of the information is redundant across the modules • The last two modules aimed to cover too many topics • More training on digital transformation tools for teaching • Methods to improve pedagogical resources for too many students in a classroom • Focusing on fields other than language teaching, such as mechanical engineering and robotics, would be beneficial
Teaching approach	• The collaborative tasks with other colleagues were challenging due to the lack of time to meet with them • More time to be dedicated to practical activities such as group discussions/activities

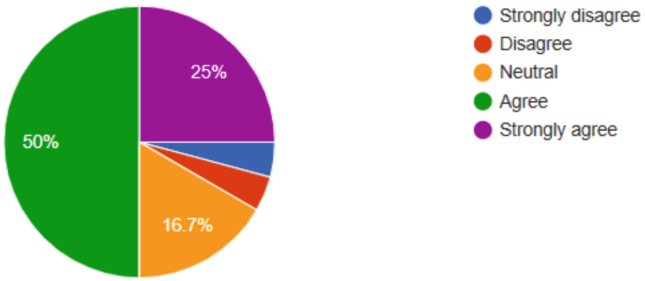


Fig. 3. The TDP4HE Training programme content met my expectations.

Also, regarding whether the synchronous sessions were organised to allow all participants to participate fully, 83.4% strongly agreed or agreed with the statement, while a few remained neutral or strongly disagreed (Fig. 4).

5.4 Content/Materials/Tools/Moodle Platform

Concerning the content and whether it was clear and understandable, 95.9% of participants strongly agreed or agreed with the statement, while one remained neutral (Fig. 5).

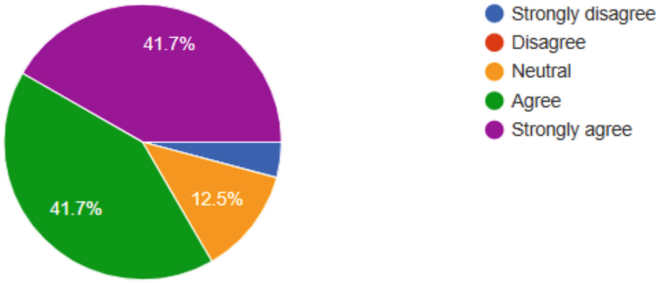


Fig. 4. The synchronous sessions were organised to allow all participants to fully participate.

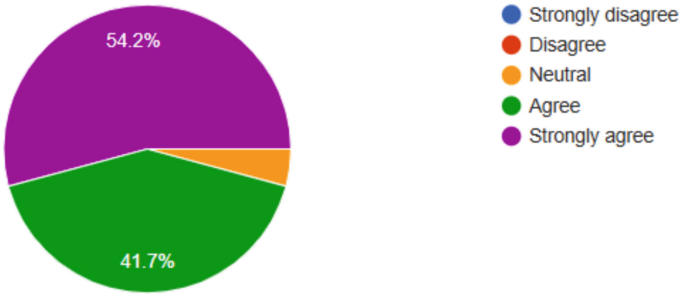


Fig. 5. The content has been clear and understandable.

Also, the organisation of content, along with the use of the Moodle platform, were appreciated. Participants valued the availability of further readings, presentations, and video recordings, which allowed them to revisit and deepen their understanding of the material.

The following excerpts highlight this theme:

I want to underline the fact that I really enjoyed the way you organised your content and how you used the Moodle platform with underline the fact that I really enjoyed the way you organised your content and how you used the Moodle platform with all those topics connected and the details (Participant 1).

So I don't know if you have the tracks on how many times people check your platform, but I have to say that I was interested in here and there to go back to one topic or to try to find out more about something else (Participant 1).

Results from the questionnaire report that the LMS/ Moodle platform worked well (Fig. 6), 83.4% of participants strongly agreed or agreed with the statement, while a few remained neutral or strongly disagreed.

The focus group discussion also highlighted the value of the online format and the use of interactive tools like breakout rooms. Participants found the programme's flexibility and the ability to engage in meaningful discussions with peers beneficial.

The excerpts below showcase this theme:

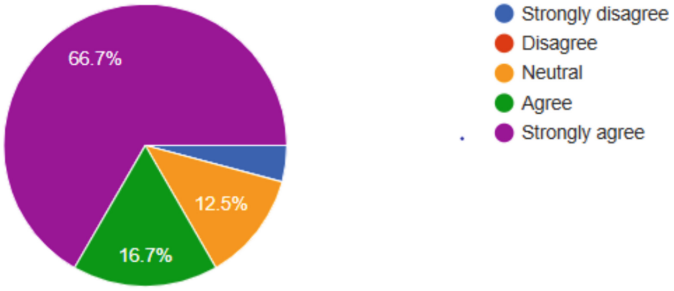


Fig. 6. The LMS/ Moodle platform worked well.

In this way, it's very useful to break the groups into different groups, smaller groups. I think that this has been a very good idea from Patrick, and I think that we can do it more frequently (Participant 4).

I think it was really nice to keep this combination of the plenaries, the sessions, the online sessions, and the interaction in the breakout rooms (Participant 1).

Regarding whether the pre- and post-activities were useful, 79.2% of participants strongly agreed or agreed with the statement, while a few remained neutral or strongly disagreed (Fig. 7).

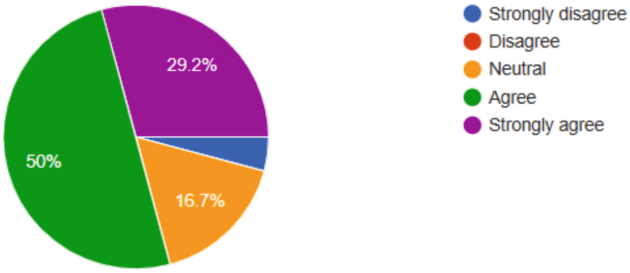


Fig. 7. The pre- and post-activities were useful

The integration of technology was another aspect of the content highlighted in the programme. Participants recognised the need to move beyond basic digital tools to more advanced technologies like virtual reality to enhance student engagement and motivation.

The excerpts below illustrate this theme:

I would say that nowadays in the 21st century we can't speak about technology only using, for example, a presentation or a video in our classroom, just to use technology for the sake of technology. So it was very insightful that we had also a module concerning the virtual reality and all of its aspects (Participant 3).

So you can see different ideas how you could improve yourself and your approaches, which was very useful from my personal perspective. And of course, the digital, the use of digital tools is very important, just because the students, they live in this kind of environment, so we have to try to be closer to them, to try to, I know, a part of their lives, so the education won't be just something they have to pass to get the degree or

something like that, to involve them really in as many activities as possible. So technology is different tools, it provides these possibilities. (Participant 2).

I want to underline the fact that I really enjoyed the way you organized your content and how you used the Moodle platform with underline the fact that I really enjoyed the way you organized your content and how you used the Moodle platform with all those topics connected and the details the details about the further readings and the PowerPoint presentations and the video recording of your presentation (Participant 1)

5.5 Modules

Regarding the question of whether the training programme modules were helpful so that they are more confident about its topic now, most participants strongly agreed or agreed with the statement for all modules (Fig. 8). Participants tend to strongly agree or agree about being more confident for Modules 2 and 5. On the other hand, Modules 1 and 4 concentrated some neutral responses. Only one participant strongly disagreed with the statement on all occasions.

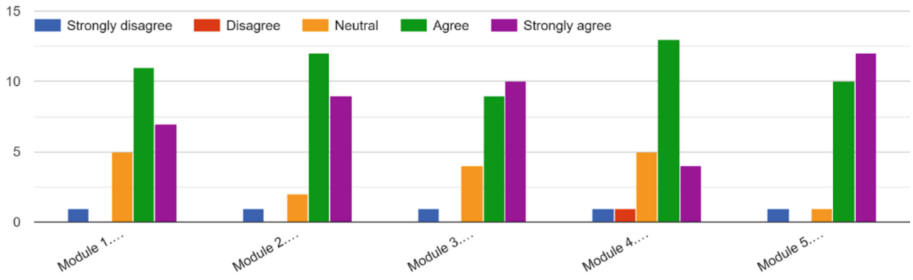


Fig. 8. The TDP4HE modules were useful, so that I am more confident about its topic now.

They would also recommend this training programme to other colleagues, as 75% of them strongly agreed or agreed with the statement, while a few remained neutral (Fig. 9).

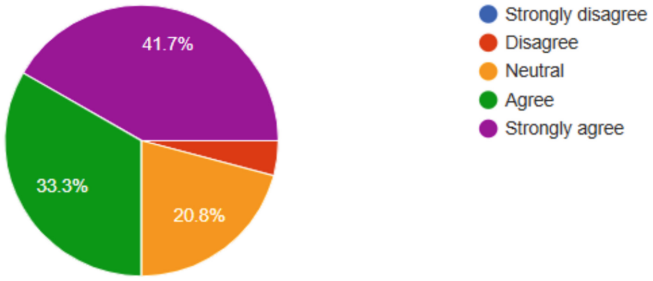


Fig. 9. I would recommend this training programme to other colleagues.

5.6 Challenges Faced

Despite acknowledging the importance of technology, participants pointed out significant challenges, including financial constraints and the complexity of integrating advanced technologies into their teaching practices:

Of course, we talked about any challenges that we may have about the equipment and that it costs a lot of money. However, it is very important to face these challenges and integrate technology in our classroom and not just a presentation or a video (Participant 3). […] *Yeah, I totally agree with you, [the participant calls the name of another participant here]* […] *And again, right, I agree with this aspect about the, I mean, with the financial aspect that it might be challenging in all institutions too.* (Participant 1).

5.7 Suggestions for Changes/Improvements

Participants suggested several improvements, including more active engagement with the underutilised community of practice to enhance course development and discussions despite time constraints. Future changes could involve reorganising the programme with additional or different partners and incorporating workshops or small conferences to enrich face-to-face interactions. Additionally, there is a need for more relevant examples and modules focused on designing virtual or augmented reality tools tailored to various specializations and scenarios, as current examples may not suit all teaching contexts.

The excerpts below showcase this theme:

I think that the community of practice is a tool that we have to use and we are not using it. Because many of the ideas, many of the things, many of the thoughts that we have covered in the course can be developed and discussed in the community of practice. We have to feed the community of practice. I think that it's not easy because we have a lack of time for doing so many things, but I think that we have to encourage ourselves and encourage our colleagues to use the community of practice. I think that it's just to ignite it. So, we start interchanging ideas in the community of practice. This is one of the things that I think could improve a lot the courses and our regular teaching because it is with the interaction, it is with the change of ideas, how this thing can flourish (Participant 4).

My first suggestion is to do it again next year maybe to involve more partners or to separate different activities to different partners so everyone will be involved in these activities that's just suggestion of course and the second thing I think it will be very useful if we have a workshop or a small conference or something like that and we can meet in person. I mean divided in different groups of different tasks. I think that the work face to face could never be the same. I mean, it's much more useful and enrich all the people participating in it (Participant 2).

I really enjoyed the one with Dali that you had as an example included in your course, but I'm not sure of how relevant that is in my teaching context. So maybe one extra module that you could include or at least one session would be to create those groups groups in which we might design one particular virtual reality tool or augmented reality tool that could work or not, I'm not sure. But I felt at that stage of the module that further into specific scenarios, cases for different specializations (Participant 1).

But in the two sessions I attended, what I miss what's more focus in specific problems. Let's say, to use technology is very far from easy. When you want to integrate, for example, virtual reality in your class is not something that you can do easily. You need, I think, I think my experience, at least in the things that I am teaching, for me is not easy. And there is another thing that I miss is how to provide interaction with technology. (Participant 4)

5.8 Recommendations

Based on the results of the focus group and the evaluation questionnaire, several recommendations can be proposed.

Firstly, it would be beneficial to engage participants before the programme begins to maintain and boost initial curiosity and motivation.

In terms of pedagogical validation and reassurance, the programme could incorporate case studies or testimonials from educators who have successfully implemented similar methodologies. This not only provides practical validation but also inspires participants by showcasing real-world applications. Furthermore, hosting interactive Q&A sessions where participants can discuss their existing methods with experts and receive constructive feedback can be highly beneficial. This approach reinforces the validity of their practices while offering opportunities for improvement.

Participants would also be interested in more practical approaches, particularly in handling large class sizes, which would improve the relevance of university teaching contexts.

To address the theme of integrating technology, the programme should include hands-on workshops that focus on using advanced technologies like virtual reality, augmented reality, and AI tools in teaching. These workshops can provide practical experience and confidence in using new tools. Additionally, providing comprehensive guides and tutorials tailored to various teaching contexts can support participants in effectively integrating these technologies into their classrooms.

Recognising the challenges in implementing technology, it is crucial to discuss strategies for resource allocation, such as acquiring funding or grants to overcome financial constraints. Simplifying the integration process by offering step-by-step guides can also help participants incorporate technology with minimal complexity and resources. This dual approach addresses both financial and practical barriers.

The programme's collaborative learning and networking aspect can be enhanced by organising participants into smaller, themed groups based on their disciplines or interests. This facilitates more relevant and in-depth discussions. Additionally, planning periodic online meetups can help maintain and strengthen professional connections, fostering a robust network of educators. The TDP4HE Community of Practice can assist in this effort.

Enhancing the interactive and flexible learning environment is another key area for improvement. Providing flexible scheduling options can also accommodate participants' diverse needs and time zones, ensuring everyone can participate fully.

Regarding suggestions for improvement, it is advised to make greater use of the underutilised TDP4HE Community of Practice, which can significantly enhance discussions despite time constraints. Furthermore, there is a need to integrate more relevant examples and modules that focus on designing virtual or augmented reality tools specific

to various specialisations and scenarios, as the current examples may not apply to all teaching contexts.

To address the need for longer interaction periods, extending the time allocated for breakout room interactions can allow for more thorough discussions and exchanging ideas.

In general, the results indicate that participants appreciated the training programme's innovative approach to transformative digital pedagogies. These insights have supported refining the training strategy and addressing the identified issues to better meet educators' needs and enhance overall satisfaction with the programme.

6 Discussion

Digital transformation and the acquisition of pedagogical competence is vital, and for this reason it has been listed among the strategic priorities of the European Commission [1]. By embracing technological change and equipping faculty with the necessary skills, HE institutions can better serve their students and contribute to a more digitally literate and globally competitive workforce. Digital transformation is not just about using technology for technology's sake. It rather concerns the thoughtful and strategic integration of technology to enhance teaching and learning, prepare students for the future, and drive innovation in HE. This task requires a significant investment in faculty development to ensure that academics have the support they need in order to develop the pedagogical competence to effectively utilise technology in their teaching practices. The purpose of the study was the evaluation of the pilot version of the TDP4HE training programme offered online to members of the academic teaching staff of the Universities of the European University of Technology alliance (EUt+) in Spring 2024. Through the administration of an online questionnaire and the organisation of semi-structured focus groups, useful results were obtained in relation to participants' perceptions about the programme, challenges faced during its implementation and suggestions and recommendations for future improvements.

First and foremost, the study revealed a common characteristic in all the participants; this was the fact that they all recognised how important it was to develop their pedagogical competences in combination with the integral use of digital technologies dur-ing the teaching and learning process. This is found in literature in many different contexts [16, 24]. According to Bitar and Davidovich [14], educators are prompted to rethink their instructional roles and enhance their teaching strategies, placing emphasis on continuous professional growth.

In relation to participants' perceptions about the training programme, this was positively perceived by the participants, who characterised it as useful for their profes-sional training. In general, participants spoke highly of the training programme, stat-ing that it met their expectations, and that they were satisfied with the platform use, the organisation, the material as well as the content. In addition, participants appreciated the fact that the trainers had diverse backgrounds, since this enriched their learning experience. Moreover, the selected topics and the fact that theory was combined with practice and that there were opportunities for communication and exchange of ideas were viewed positively by the participants. Bragg et al. [25] support that online professional development

for teachers is more productive than traditional face-to-face professional development when it is relevant to teachers' professional learning needs promotes social aspects of learning includes content that is readily transferred to classroom environments, and when it encourages self-reflection. Participants' positive comments in the present study pertaining to the content of the training programme show that they valued the aspects of transformative digital pedagogies covered in the training, and that the fact that the programme was well-structured and well-organised on the Moodle platform facilitated this process.

As far as challenges encountered are concerned, participants reported financial constraints and the complexity of integrating advanced technologies into their teaching practices. These kinds of constraints and lack of appropriate equipment is a challenge faced in HE in many parts of the globe, as illustrated in the systematic literature review by Forde and O'Brien [26].

Furthermore, the study also yielded insightful results with regards to improvements that need to be made in the second cycle of the TDP4HE training programme implementation. These suggestions mainly related to the inclusion of more interactive and practise-based activities during the training, which is in line with ElSayary's [27] view that it is essential to provide teachers with realistic experiences for integrating digital technologies into the teaching and learning process. According to ElSayary [27], training should focus on collaborative work to reduce anxiety and create opportunities for exchange of good practices, and this is in fact what the TDP4HE training programme aimed to offer. Nevertheless, participants stated that despite the opportunities for interaction and discussion and collaboration in breakout rooms during the synchronous webinars, they still needed less theoretical presentations and more time for interaction and practice during the training.

The evaluation of the TDP4HE training programme yielded important insights for the improvement of the programme and its re-implementation with a wider audience, outside the bounds of EUt+. The study also carried important implications for HE institutions, teacher educators, academics and researchers in the field of digital trans-formation and education in general. The iterative process of DBR, on which this study is based, is hoped to deliver richer results that could further facilitate the extraction of more solid conclusions and could inspire further research on the topic.

7 Conclusion

In conclusion, the evaluation of the Transformative Digital Pedagogies for Higher Education (TDP4HE) training programme has provided valuable insights into participants' perceptions, highlighting both its strengths and areas for improvement. The programme represents a significant step towards advancing digital pedagogies in higher education by equipping academic staff with the necessary skills to effectively integrate technology into their teaching practices. The findings from this study underscore the importance of aligning professional development initiatives with the real-world challenges and expectations of educators. By addressing participant feedback, future iterations of the programme can enhance its impact by incorporating more hands-on activities, fostering greater interaction among participants, and expanding the range of technological tools

covered. Furthermore, the recommendations put forth highlight the need for sustainable support structures, including access to funding opportunities, dedicated mentorship, and continuous engagement through professional networks like the TDP4HE Community of Practice. Implementing these improvements will ensure that the programme not only remains relevant but also becomes a model for digital transformation in higher education. Ultimately, by fostering a culture of innovation and collaboration, the programme has the potential to empower educators, enhance student learning experiences, and contribute to a more digitally literate and globally competitive academic landscape. The next implementation of the programme is expected to build upon these insights by offering an even more refined and interactive learning experience, reaching a wider audience beyond the EUt+ alliance and further solidifying its role as a key driver of digital pedagogical transformation. As research suggests, professional development programmes that effectively integrate digital pedagogies, provide hands-on experiences, and encourage collaboration are essential for fostering meaningful educational innovation [25].

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